

Mobile Application Development with Flutter

Lecture 42. Switch, CheckBox and Radio Button in Flutter

Introduction

- How do users select options in apps?
- Examples:
 - Enable/disable setting → Switch
 - Select multiple options → Checkbox
 - Select one option → Radio
- These widgets are used to take user input and control app behavior

Common Examples

- Turn notifications on/off
- Accept terms and conditions
- Choose gender
- Select hobbies
- Choose one payment method

Prepared by Ehtisham Rasheed

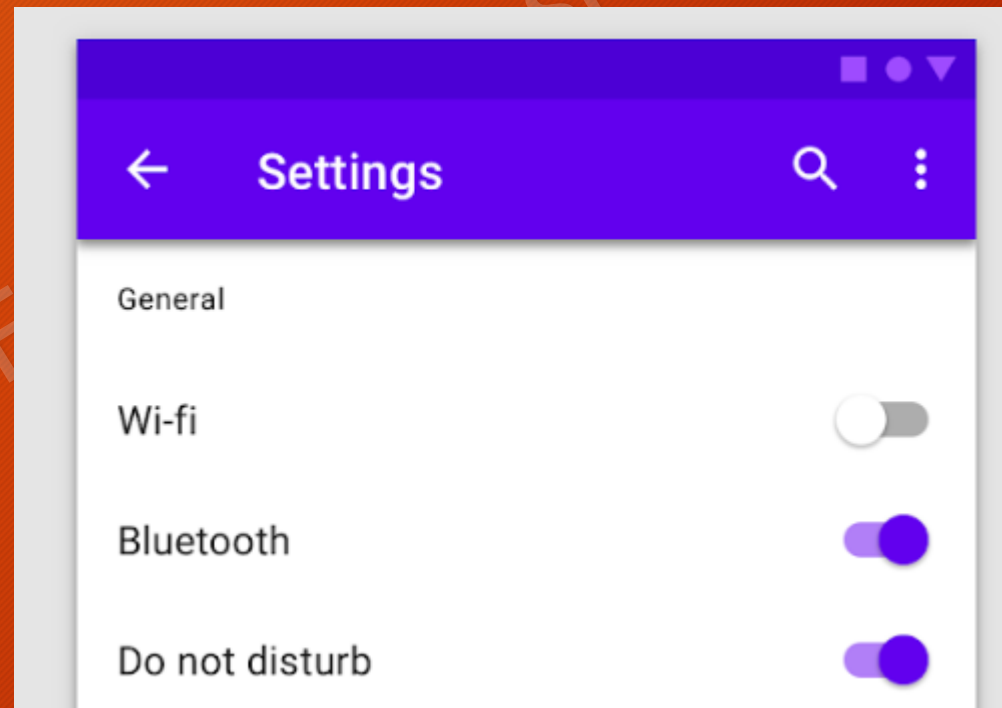
1. What is Switch?

- A Switch is used for true/false or on/off type input

- Real-World Examples

- WiFi ON/OFF
- Dark Mode
- Notifications

- Use it when the user is controlling a setting that has only two states

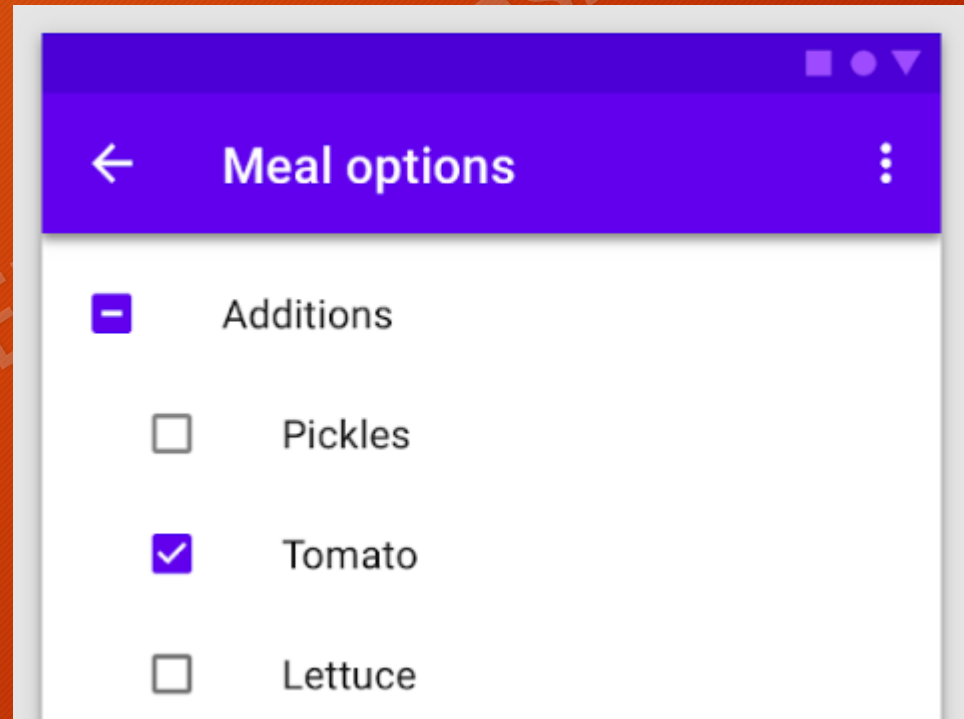


SwitchListTile

- A **SwitchListTile** is a "convenience widget" that combines a **Switch** with a **ListTile**
- In a standard **Switch**, we only get the toggle button. If we want a label (like "Notifications") next to it, you have to manually build a **Row** with a **Text** widget.
- **SwitchListTile** does all that layout work for us and makes the **entire row** clickable

2. What is Checkbox?

- Checkbox allows selecting multiple options
-  Real-World Examples
 - Accept Terms & Conditions
 - Select multiple subjects
 - Filters
 - Select hobbies

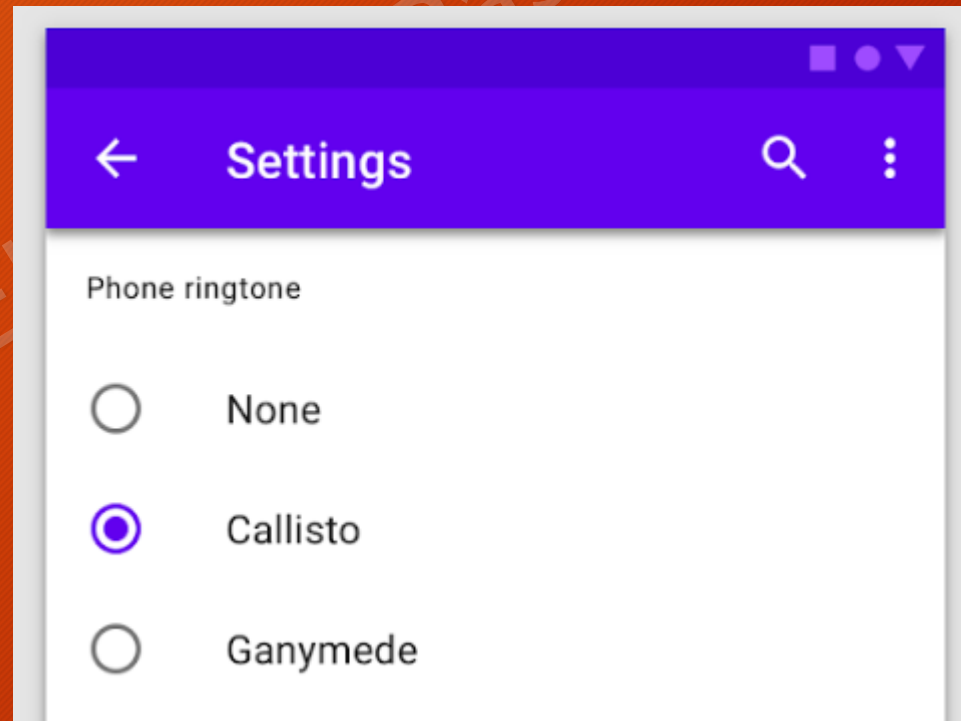


CheckboxListTile

- CheckboxListTile combines
 - ListTile
 - CheckBox
- The entire tile is interactive, not just the checkbox itself
- Use it when you want labeled checkbox options in a neat form layout

3. What is Radio Button?

- Radio buttons allow selecting ONLY ONE option from a group
-  Real-World Examples
 - Gender selection
 - Payment method
 - Quiz answer
 - Delivery option



Important correction

```
Radio(  
  value: "Male",  
  groupValue: selectedValue,  
  onChanged: ...  
)
```

- But groupValue and onChanged on Radio are now deprecated. Flutter recommends using RadioGroup to manage the selected value for the radios in its subtree

New Radio Approach: RadioGroup

- `RadioGroup<T>` manages the selected value for all radios of the same type inside it.
- It also improves keyboard navigation and semantics

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